

## Appendix 2

Table 2A: List of selected papers, with their allocated themes, the key findings and methodology  
Themes: T1= Attainment, T2= Affect, T3=Participation, T4=Identity, T5=Intersectionality

| AA       | Reference                 | T1 | T2 | T3 | T4 | T5 | Key finding (For gender)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Level of Education | Methodology  |
|----------|---------------------------|----|----|----|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------|
| Proced2  | Kyeleve & Williams (1995) |    | •  |    |    |    | Mixed (Attitudes). Boys > Girls in enjoyment of group work, technology, and confidence; Girls > Boys in enjoyment of seeking clarification and lower anxiety on tables/graphs.                                                                                                                                                                                                                                                                                                                                          | Post-16            | quantitative |
| Proced3  | Parsons (1995)            |    |    |    | •  |    | F = Collaborative / M = Competitive. All-female groups favored a supportive "jigsaw" style where everyone contributed, whereas mixed and all-male groups were often dominated by individualistic, competitive male behaviors that marginalized females or less experienced peers.                                                                                                                                                                                                                                       | Higher Education   | qualitative  |
| Proced7  | Woodrow & Jarvis (2001)   |    |    | •  |    | •  | M > F (Math/Informatics Participation). Men are significantly more likely to study math and informatics across all ethnic groups (e.g., Asian Men 33% vs Women 12%). Conversely, women are three times more likely to study languages than men                                                                                                                                                                                                                                                                          | Higher Education   | quantitative |
| Proced10 | Mendick (2001)            |    |    | •  | •  |    | F = Gender Category Maintenance Work. Girls often reject mathematics as a way to perform and maintain their feminine identity, as math is socially coded as a masculine indicator of elitism and intelligence.                                                                                                                                                                                                                                                                                                          | secondary          | qualitative  |
| Proced11 | Mendick (2004)            |    |    |    | •  |    | M > F (Identification). Based on author's PhD data (n=43), 6 out of 24 boys identified strongly as "math people" compared to 0 out of 19 girls. Mathematics is culturally associated with "separation" and "masculinity," while femininity is linked to "connection" and "emotion," creating a conflict for girls.                                                                                                                                                                                                      | secondary          | qualitative  |
| Proced13 | Anghileri (2005)          | •  |    |    |    |    | M > F (Attainment & Mental Strategies). Boys (mean 5.3) significantly outperformed girls (mean 4.4, $p < 0.005$ ) in 2003. Boys used informal methods (35%) and mental strategies (12%) more frequently and successfully. Girls relied more on standard written algorithms (35%) and achieved better results in schools where they successfully used the 'chunking' algorithm to organize their work.                                                                                                                   | primary            | quantitative |
| Proced14 | Jay (2005)                | •  |    |    |    |    | M > F (Improvement & Flexibility). Boys improved significantly more than girls across trials <sup>21</sup> . For girls, improvement is strongly predicted by the appropriateness of the strategy (use of steepness/gradient) used in the first trial ( $\beta=0.044$ ), indicating a preference for strategy consistency <sup>22</sup> . In contrast, boys' improvement is more flexible and is marginally predicted by the presentation order of sparse-to-realistic contexts rather than initial strategy persistence | secondary          | quantitative |
| Proced15 | Mendick                   |    |    |    | •  | •  | Media Binaries (Reason vs. Calculation). The "Vorderman factor" significantly increases the number of                                                                                                                                                                                                                                                                                                                                                                                                                   |                    | Theore       |

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|          | (2005)                       |   |   |  |   |   | girls drawing female mathematicians (56.9% in UK vs. 30.5% in US), but her image as a "calculator" rather than a "reasoner" still maintains gendered binaries. Female geeks (like Willow) must be redeemed by the "heterosexual gaze" to enter adulthood, whereas male geeks (like Seth Cohen) are celebrated as "chic". Subversive slogans like "I'm too pretty to do math" are used by female students to playfully deconstruct harmful stereotypes.                              |                  | tical         |
| Proced16 | Rodd & Brown (2006)          |   |   |  | • | • | Critique of Biological Essentialism. The authors challenge the hard-wired dichotomy using data (n=93) showing that students of both genders exhibit hybrid orientations of systemizing and empathizing. They highlight that Systemizing Quotient (SQ) tests are biased toward male-centric domains (e.g., electrical wiring), while women's systemizing in traditionally feminized domains (e.g., domestic management or cooking) is systematically excluded or rendered invisible. | Higher Education | Theore tical  |
| Proced17 | George (2007)                | • |   |  | • | • | Math as "Manhood" & Track-based Disadvantage. Interviews reveal an implicit belief that being good at math is a male characteristic ("what kind of man..."), while for mothers, failing math is normalized. Critically, in single-sex middle-class schools, girls are disadvantaged by math-specific ability grouping that tracks them into "Basic Proficiency," effectively blocking female-aspirational career paths like accounting                                              | secondary        | mixed         |
| Proced18 | Diego-Mantecon et al. (2007) |   | • |  |   |   | M > F (Competence & Enjoyment) / F > M (Rejection of Rote). Girls report significantly lower enjoyment and lower personal/relative competence than boys. However, girls are more progressive in their epistemological view, being more likely to reject the "rote knowledge" myth than boys.                                                                                                                                                                                        | secondary        | quantit ative |
| Proced19 | Little (2007)                | • | • |  |   |   | F > M (Positivity & Persistence). Girls were significantly more positive toward coursework than boys (77% vs 64%). Girls invested 29% more time (9.41 hours vs 7.28 hours for boys). Although girls perceived the task as more difficult, the author interprets this as evidence of higher engagement and persistence rather than lower ability.                                                                                                                                    | Post-16          | mixed         |
| Proced20 | Yu (2008)                    |   | • |  |   | • | No Diff (Teacher Beliefs). Gender does not in general affect teachers' beliefs about math teaching, the nature of math, or the purposes of math education in either England or China.                                                                                                                                                                                                                                                                                               | secondary        | mixed         |
| Proced21 | Hodgen et al. (2009)         | • | • |  |   |   | F < M (Confidence & Enjoyment). Girls' enjoyment of math lessons drops more sharply than boys' by age 14 (50% vs 59%). Most critically, by age 14, only 52% of girls believe they are "good at maths" compared to 81% of boys, despite only marginal differences in actual attainment. Fewer girls (26%) intend to pursue mathematics post-16 compared to boys (35%).                                                                                                               | secondary        | mixed         |
| Proced22 | Pampaka & Williams (2010)    | • | • |  |   |   | Mixed / Small DIF (Self-efficacy). No simple gender gap in mathematics self-efficacy is reported. Item-level DIF shows statistics items favouring female students, while modelling and ratio/proportion items favour male students; course-group differences                                                                                                                                                                                                                        | Higher Education | quantit ative |

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|          |                            |   |   |   |   |   | are more central than gender.                                                                                                                                                                                                                                                                                                                                                                                                                       |               |              |
| Proced23 | Meremikwu & Enufoha (2010) | • |   |   |   | • | No Diff (General Attainment). Gender does not have a significant main effect on mathematics achievement. However, interaction effects show that female pupils in private schools taught with instructional aids achieved significantly higher scores than their male counterparts.                                                                                                                                                                  | primary       | quantitative |
| Proced25 | Onion (2011)               |   |   |   | • | • | F = Narrative Disidentification / Classed Memory. Women across three generations narrate low confidence and negative school mathematics experiences; poorer background, teacher feedback and family-based comparisons shape long-term disidentification from mathematics.                                                                                                                                                                           | non-education | qualitative  |
| Proced27 | Deacon & Edwards (2012)    |   | • |   | • |   | F < M (Confidence in Support). Females reported less confidence in their capacity to motivate or support their peers than males. Additionally, while females showed more consistent responses in group work preferences, they expressed higher uncertainty about the specific sources of their own mathematical motivation compared to male students.                                                                                               | secondary     | mixed        |
| Proced28 | Partanen & Kaasila (2012)  |   |   |   | • |   | M > F (Interactional Dominance). Males exhibit assertive styles, including interrupting and "posturing," which prohibits females from expressing mathematical ideas. Females use egalitarian strategies (e.g., tag questions, conditional forms) to avoid appearing as authorities, which makes their contributions vulnerable to being overwritten by male dominance.                                                                              | secondary     | qualitative  |
| Proced29 | Adams (2014)               |   |   |   | • |   | F = Professional Isolation / Agency. Women mathematics teachers describe constrained access to mathematics-focused professional learning, limited agency and isolation in school; collaborative professional communities support stronger professional learning identities.                                                                                                                                                                         | secondary     | qualitative  |
| Proced31 | Quaye (2014)               | • | • |   |   | • | No Clear Gender Effect / Class and Habitus Stronger. Gender is modelled alongside social class and ethnicity; the reported achievement patterns are better explained by class, cultural capital and mathematics-related habitus than by gender alone.                                                                                                                                                                                               | secondary     | quantitative |
| Proced33 | Smith & Golding (2015)     | • |   | • | • |   | F = Performance / Participation (Good Practice). Success in recruiting girls is driven by teachers who explicitly and repeatedly confirm that girls are capable of succeeding at A-level. Critical factors include providing early evidence of success through "extended curricula" (demanding math before age 16) and framing mathematics as a versatile tool that "opens doors" to a wide array of interests rather than a narrow elite pipeline. | secondary     | qualitative  |
| Proced35 | Pomeroy (2015)             |   |   |   | • | • | M = Physical Masculinity / Disidentification. A form of non-mathematical masculinity, tied to class and ethnicity, positions academic mathematics as incompatible with valued masculine identities.                                                                                                                                                                                                                                                 | secondary     | qualitative  |
| Proced36 | Alam (2015)                |   |   | • |   | • | F = Performance / Participation (Equal Domination). Girls reject the stereotypical                                                                                                                                                                                                                                                                                                                                                                  | secondary     | mixed        |

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|          |                                      |   |   |   |   |   | belief that boys have higher innate ability in math. However, actual participation in 'Higher Maths' is low due to perceived difficulty, a lack of clear interest, and the necessity of external financial resources (private tutoring).                                                                                                                                                                                  |           |              |
| Proced37 | Foley (2015)                         |   | • |   | • |   | F = Internalized Responsibility. Unlike the "fixed mindset" often attributed to girls, these participants showed a productive disposition, believing effort leads to success. However, this leads to an intense sense of individual responsibility; girls feel personal guilt and worry when they do not make expected progress, viewing failure as a personal shortcoming rather than a structural or pedagogical issue. | primary   | qualitative  |
| Proced38 | Spencer & Fielding (2015)            | • |   |   |   |   | F > M (Tool Adoption in Upper Primary). While gender differences in pictorial use were evident in the pre-test (girls used a wider range of drawings), the post-test showed a dramatic split by age: in Year 6, 81% of girls adopted the Bar model compared to only 46% of boys. This suggests that female students may be more receptive to structured visual modeling as a means to manage mathematical complexity.     | primary   | mixed        |
| Proced39 | Alam (2016)                          |   | • |   |   | • | M > F (Confidence). Statistical analysis shows no significant gender difference in overall attitude scores. However, boys score significantly higher on the "Confidence" sub-scale. Furthermore, girls are significantly more likely than boys to view mathematics as a "male domain".                                                                                                                                    | secondary | quantitative |
| Proced41 | Cascella, Pampaka, & Williams (2018) | • |   |   |   | • | M > F (Attainment). Boys consistently outperform girls, with the national advantage being approximately 7 points. However, this gap is highly sensitive to locality and school track: gender disparities increase moving from North to South Italy and are notably stronger in vocational schools (Professionali) compared to academic high schools (Licei).                                                              | secondary | quantitative |
| Proced42 | Lei & Pampaka (2018)                 |   | • |   |   | • | M > F (Self-efficacy). Male students exhibit significantly higher self-efficacy scores than female students at every secondary grade level. Critically, while male confidence shows a cumulative upward trend, UK female students report an "obvious drop" in MSE precisely between Year 8 and Year 9 (age 13-14), a dip not observed in their male peers.                                                                | secondary | quantitative |
| Proced43 | Lord (2020)                          |   |   | • |   |   | F = Performance / Participation (Altruistic Disconnect). Despite being high-attaining in A-level mathematics, females rejected math degrees because of a perceived disconnect between the subject and their altruistic career aspirations. They narrate mathematics as an isolated or purely technical field that does not allow for the "helping others" identity they seek.                                             | Post-16   | qualitative  |
| Proced44 | Mendick et al. (2020)                |   |   |   | • | • | M = Misogynistic Revenge / F = Empathic Resilience. Contemporary popular culture constructs a "geek" masculinity where mathematical brilliance justifies a narrative of suffering and subsequent revenge against women. This narrative legitimates the policing of geek spaces. In contrast, female geek representations like Betty Aurora Rincón demonstrate that suffering can lead to empathy, providing a             |           | qualitative  |

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|          |                               |   |   |   |   |   | discursive counter-model to the dominant "toxic nerd" identity.                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |              |
| Proced45 | Abbott (2021)                 | • |   |   |   | • | F > M (Performance & Perception). Following the instructional intervention, female students not only achieved higher post-test scores than males but also expressed a significantly stronger preference for this structured instructional method, suggesting the worked example effect is more prevalent among female learners in this context                                                                                                                                                            | secondary          | mixed        |
| Proced46 | Golding et al. (2021)         | • | • |   |   |   | M > F (Self-efficacy) / F > M (Anxiety). Female students consistently report significantly lower self-efficacy (belief in their ability to solve specific math problems) and significantly higher mathematics anxiety compared to males.                                                                                                                                                                                                                                                                  | secondary          | quantitative |
| Proced48 | Boli & Golding (2024)         | • | • | • |   |   | F > M (Progress/Engagement) / M > F (Stagnant performance). Males are over-represented in categories where grades remain stagnant or deteriorate during resit attempts. Females often exhibit higher levels of mathematics-related anxiety, yet they demonstrate different, sometimes more persistent, patterns of engagement with complex word problems compared to their male peers.                                                                                                                    | Post-16            | mixed        |
| Proced49 | Kerrison (2025)               | • |   |   |   | • | No Diff (Initial Performance). While the study's initial data collection attended to gender, it reported that the difficulty with real-life context problems (like speed and distance) was a generalized issue across the cohort. Socioeconomic background (SES) and associated cultural capital were identified as more critical variables than gender in determining success in these specific mathematical tasks.                                                                                      | secondary          | mixed        |
| Proced50 | Golding & Wilkinson (2025)    | • | • | • |   |   | M > F (Attainment & Affect). England's gender gap in mathematics achievement has reopened significantly, with boys outperforming girls in Year 9 by the widest margin globally. This attainment gap is preceded by a "confidence and enjoyment" gap that is already distinct in Year 5, alongside gendered career aspirations.                                                                                                                                                                            | primary, secondary | quantitative |
| Proced51 | Logan (2025)                  | • |   | • |   |   | F > M (Exceptional Attainment). Quantitative data shows that in subjects where they are the minority (math, physics, computing), female students achieve significantly higher 'A' grade pass rates than their male counterparts. Qualitative data suggests that for girls, exceptional attainment is a "social prerequisite" to enter and "fit in" to competitive, male-dominated mathematical spaces.                                                                                                    | Post-16            | mixed        |
| Proced52 | Compton & Unwin-Berrey (2018) |   | • |   | • |   | F = Emotional Closeness / M = Instructional Distance. Boys were more likely to draw the teacher at the board or desk (distant), focusing on the "instructional" aspect. In contrast, girls more frequently drew the teacher among the pupils, highlighting "emotional closeness" and praise. Furthermore, girls were nearly twice as likely as boys to characterize mathematics as "difficult," whereas boys were more likely to label it as "fun," despite 10 instances of boys labeling it as "boring." | primary            | mixed        |